Thomas Chisica Londoño

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Research Profile

Applied mathematics student working on AI evaluation and measurement validity: replicating capability measurements, LLM-judge stability, and RL policies tested against calibrated baselines, with held-out evaluation and explicit limits on what results show.

Education

Universidad del Rosario, Bogotá 2024–present

BSc in Applied Mathematics and Computer Science. Expected graduation: June 2028.
GPA: 4.11/5.00 | Full-tuition merit scholarship.

Relevant coursework: Machine Learning, Probability & Statistics, Real Analysis, Linear Algebra, Graph Theory, Differential Equations, Optimisation and Algorithms & Data Structures.

Research Experience

AI evaluation and measurement validity 2026–present

Independent projects

- Replicated METR's Time Horizon 1.1 with its code and data (all 20 public models exact; 128.7-day doubling). With only measured human anchors (162 of 228 tasks) the frontier horizon is not identified, while the doubling time barely moves (unpublished pilot).
- On 120 Petri calibration transcripts, three LLM judges reached ICC 0.74–0.88, yet on misuse cooperation (ICC 0.81) the target order held under all judges in only 34% of bootstrap draws; found silent scoring defects in Petri. Main study pre-registered. Pilot

Simulation and reinforcement learning for supply-chain resilience 2025–present

Undergraduate research with Prof. Alexander Garrido, Universidad del Rosario

- Reconstructed Garrido-Ríos's (2017) thirteen-operation supply-chain model in SimPy with Gymnasium environments for PPO, tests and source-model checks.
- Benchmarked 10 PPO checkpoints on 60 held-out streams against a calibration-selected static policy under the same decision contract; PPO trailed it (ahead in 2 of 10 checkpoint means), so the apparent RL advantage was withdrawn.

SCRES: simulation and evaluation code

Collective behaviour and human coordination 2025–present

Independent reanalysis; software for Prof. Edgar Andrade-Lotero's lab, Universidad del Rosario

- Independent reanalysis of Andrade-Lotero and Goldstone's (2021) division-of-labour experiment: modelled the 8×8 board as a graph, reconstructed past-only features for 45 dyads and checked temporal integrity with six tests.
- Certified the graph's exact minimum conductance (22/210, attained only by the four half-board cuts) and implemented basis-invariant Fiedler-subspace projectors. Spectral and coordinate energies were nearly redundant; predictive advantage remains unestablished.
- Simulated identifiability: in table form the rival models were not identifiable with this design; parametric rewrites helped, but not for every parameter.
- For the lab's PsyNet experiments, wrote docs and smoke tests and built tutorials, task maps and a score screen for the foraging_trucks demo (merged October 2026).

Spectral/SODCL: reanalysis and temporal audit

Recurrent neural networks: connectivity and motor learning July–October 2026

Neuromatch Academy Computational Neuroscience (68 h): team project and independent follow-up

- Ran Python/NumPy experiments on 200-unit FORCE-trained recurrent networks across four connec-

tivity densities and 16 paired seeds, with learning disabled during held-out evaluation.
– Added an equal-plasticity control: with the same trainable-edge budget at every density, the density advantage disappears (16/16 seeds). An earlier 8-seed reversal did not hold up with more seeds.

Motor-RNN experiments and equal-plasticity control

Animal social networks: yellow-bellied marmots 2026–present

Research with Dr. Adriana Maldonado-Chaparro, Universidad del Rosario

– Built observation-aware SQLite dyadic tables (interactions, sightings, residence, relatedness); exploratory count/mixed-effects models and matched samples to examine sampling effects.
– Validated a legacy NetLogo/R individual-based model with Python and native-runtime checks; found homophily-rule defects and documented reproducibility limits.

Technical Skills

Programming & ML: Python, PyTorch, Gymnasium; C, R, SQL.
Scientific computing: NumPy/SciPy, SimPy, NetworkX, statsmodels; NetLogo model audits.
LLM systems & software: RAG, embeddings, FAISS, FastAPI; Git, Linux, pytest/CI, SQLite, PostgreSQL, Polars, LaTeX.
Languages: Spanish (native); English C1 (IELTS General Training 7.0, 2023).

Teaching, Awards & Funding

Calculus I Teaching Assistant, Universidad del Rosario July 2024–July 2025

Led problem-solving sessions and supported first-year students.

Andrés Bello National Distinction 2024

Colombian Ministry of Education

Ranked 33rd nationally in the official distinction ranking for the 2023 Saber 11 examination (Colombia's national school-leaving examination); one of 50 national recipients. Highest-ranked recipient from Tunja, which ranked first among certified education authorities in the reported 2023 results.

Full-tuition merit scholarship, Universidad del Rosario 2024–present

Additional Experience

Witty: retrieval-augmented support prototype June–December 2025

Lead in Emerging Technologies, Digital Business Ventures; project with Progresión

FastAPI retrieval pipeline (embeddings, optional FAISS, LLM rewriting); heuristic answer/escalate routing.

Full-stack Developer, Briki February–May 2025

React frontend, Node.js backend and PostgreSQL work; bug fixes and technical pitches.

Selected Software Projects

HeliOS: educational RISC-V kernel 2025–2026

Contributed to a collaborative C kernel with task scheduling, timer interrupts, a UART shell, synchronisation primitives and heap allocation; includes QEMU/OpenSBI smoke tests. Kernel, architecture and test scripts

ContratIA Abierta: public-procurement review tools 2026

Review pipeline for Colombian SECOP open data (Polars, PostgreSQL, MongoDB, FastAPI, Dash) with interpretable signals and traceable evidence for human review. Data pipeline and review dashboard